

PENTAGON: Can Closed-Loop LLM Reasoning Achieve Autonomous Multi-Tool Cybersecurity Assessment?

Department of Computer Science, University of Dayton, USA

Abstract

We investigate whether a large language model, operating in a closed reasoning loop over real cybersecurity tools, can produce an autonomous vulnerability assessment that approaches human-expert coverage at a fraction of the time and cost. We introduce **ReasonChain**, a closed-loop architecture with three load-bearing properties: closed-loop replanning after every tool execution, cross-tool intelligence fusion through a shared knowledge bag, and target-aware planning that adapts the seed engine set to the target class. We evaluate ReasonChain through a controlled ablation study against 17 OWASP-class deliberately-vulnerable web applications, running 68 cells across four conditions (full / no-replan / no-fusion / random-order) with 10 real engines wired through SSH to a Kali Linux execution host. The closed-loop condition surfaces a **24.6-fold** increase in findings over the no-replan ablation (Wilcoxon $W=153$, $p=0.0001$; paired t excluding the DVWA outlier, $p=8.663e-28$, Cohen $d=Z$), including real CVE-class findings (e.g., CVE-2024-38476, CVE-2024-38474, CVE-2023-25690; all CVSS 9.8) discovered by the agent's NSE-script invocations. We also introduce a deterministic decision-quality annotator that labels every planner decision as correct, suboptimal, or incorrect, providing the first quantitative failure-mode taxonomy for autonomous web-app assessment agents. We release the reference implementation (MIT-licensed at github.com/eobi/reasonchain-core) together with every per-run report (PDF + JSON), the matrix CSV, the analysis notebook, and the rendered paper PDF, so that every claim in this paper is reproducible from a clean `git clone`.

Keywords.

1 Introduction

When organizations need to validate the security of their networks, they hire cybersecurity experts who manually run dozens of specialized tools — one tool to scan for open ports, another to fingerprint web technology, a third to brute-force directories, a fourth to test SQL injection, and so on. Each tool emits its own output format; the expert must read every output, mentally connect findings across tools, and decide what to try next. The process is slow (typically two to four weeks for a moderate-sized application), expensive (\$50K–\$150K per assessment), and brittle: prior work [1] reports that human testers miss roughly 73% of attack chains that require connecting evidence across more than two tools.

This paper asks a different question. Can an AI system that reasons in a **closed loop** — observing each tool's output and adapting its strategy before running the next tool, with no human in the loop — perform this task autonomously? And, critically, which components of the reasoning loop matter most for assessment quality? Understanding this has implications well beyond cybersecurity, for any domain where an AI agent must orchestrate specialized tools.

Three approaches dominate the current landscape. **Manual expert testing** is the gold standard but is gated by the 3.4-million-person global workforce shortage [2]. **Fixed automation** — pre-written scripts that fire tools in a hard-coded sequence — cannot adapt when a finding from tool #3 changes the right choice for tool #4. **LLM-assisted tools** such as PentestGPT [3] use a language model to suggest the next command, but a human still types it; the paper explicitly notes that "the human operator remains the executor." None of these systems close the loop in the sense we mean: an LLM that *plans*, *executes*, *observes*, and *re-plans* without intermediate human gating.

We introduce **ReasonChain**, an architecture with three properties whose combination — to our knowledge — has not been demonstrated in a published autonomous web-assessment system:

- **(P1) Closed-loop reasoning.** After every tool execution, the

LLM (or, in our reference baseline, a deterministic heuristic planner) receives the parsed output and re-evaluates its remaining plan. If a scan reveals an unexpected service, the system immediately adapts.

- **(P2) Cross-tool intelligence fusion.** Findings and facts from one tool inform decisions for subsequent tools through a shared knowledge bag. Open ports discovered by nmap automatically scope the NSE vulnerability scripts that nmap_vuln will run; URLs surfaced by the crawler are passed to the header probe.

- **(P3) Target-aware methodology.** The seed engine set adapts to the target class. A web target gets http_probe + url_crawler; network targets would get nmap + masscan first.

To test the contribution of each property we conduct a controlled **ablation study** with four conditions:

- **full:** all three properties enabled (the architectural claim);
- **no-replan:** ablates P1 — execute the initial plan, no re-plans;
- **no-fusion:** ablates P2 — each engine sees an empty facts bag;
- **random-order:** ablates target-aware ordering by shuffling the seed set.

Our contributions are:

1. **A reference implementation** of ReasonChain — a 600-line MIT-licensed Python codebase that wraps real CLI tools (nmap, nuclei, nikto, dalfox, sqlmap, wpscan, feroxbuster, dirsearch) either as local subprocesses or as remote engines over SSH to a Kali Linux host. The architecture is implemented in reasonchain-core; a richer commercial extension (Pentagon) is maintained separately for IP reasons.

2. **An empirical evaluation** across 17 OWASP-class targets and 72 matrix cells, conducted entirely on a single LAN with the research artifact (reports, CSVs, figures, notebook) published alongside the paper.

3. **A deterministic decision-quality annotator** that labels each planner decision as correct, suboptimal, or incorrect against four checkable feasibility rules and a findings-by-source count. This is, to our knowledge, the first reproducible failure-mode taxonomy for autonomous web-app assessment agents.

4. **Three CVSS-9.8 CVE detections** (CVE-2024-38476 Apache mod_rewrite SSRF, CVE-2024-38474, CVE-2023-25690 HTTP smuggling) surfaced live by the agent during the deep scan of an OWASP Juice Shop target, demonstrating that the architecture works on real, unmodified production-grade scanners — not on a curated benchmark.

The full reasonchain-core codebase, every per-run PDF + JSON artifact, the analysis Jupyter notebook, and this paper are available at <<https://github.com/eobi/reasonchain-core>>.

2 Related Work

ReasonChain occupies the intersection of four prior strands.

Manual penetration testing. The OWASP Testing Guide and the PTES (Penetration Testing Execution Standard) specify the methodology that human experts follow. Effectiveness is gated by the 3.4-million-person global cybersecurity workforce shortage [2]. Happe and Cito [1] characterize the failure mode of even experienced human testers: they miss multi-step attack chains because no human operator can hold the entire output of fifteen tools in working memory. ReasonChain's Facts bag is the machine-internal analogue of that operator's working memory.

Fixed automation. OWASP ZAP automation, nuclei chains, bash-orchestrated scripts (e.g., recon-ng's workflow engine), and SOAR playbooks all run dozens of tools in a fixed order. None re-plans against intermediate findings; if scan #3 surfaces a new service on a non-standard port, scan #4 still fires its pre-programmed command. This is the open-loop baseline our H1 ablation (replanning=False) reproduces.

LLM-assisted assessment. PentestGPT [3] is the closest published predecessor: an LLM proposes the next command, but a human types it and feeds the output back. The paper explicitly describes the human as "the executor." The system observes neither the actual output of execution nor any state of the target machine; the LLM's situational awareness depends entirely on what the human chooses to paste. ReasonChain closes that loop by executing autonomously over SSH and feeding the parsed output back into the planner without human intervention. We discuss our intended head-to-head comparison with PentestGPT in §8.

Cyber Reasoning Systems (CRS). DARPA's Cyber Grand Challenge (2016) and the more recent AI Cyber Challenge (AixCC, DEF CON 32, 2024) [6] funded end-to-end autonomous systems that find, exploit, and patch vulnerabilities. CRSes focus on compiled binaries and memory-corruption classes (heap-spray, return-oriented programming, type confusion). ReasonChain operates on running web applications and focuses on detection — a distinct problem space, but the underlying architectural principle (closed-loop reasoning over a tool ecosystem) is the same. AixCC's published architectures [Mayhem, ForAllSecure 2024; Trail of Bits 2024] confirm that the closed-loop paradigm scales to the binary class; our work shows it also scales to the web class with an entirely different engine pool.

Multi-agent LLM frameworks. AutoGPT, MetaGPT, BabyAGI, and SWE-Agent [Yang et al., 2024] demonstrate autonomous LLM tool use in general software-engineering and information-retrieval domains. None has been evaluated on cybersecurity assessment with the ablation rigor we apply here. Our LLMPanner reuses the same LLM-driven planning idea but specialises it to the security assessment loop and pairs it with a deterministic decision-quality annotator (§3.4) that lets us measure planner correctness rather than just task completion.

3 The ReasonChain Architecture

!Figure 1: The ReasonChain closed-loop architecture. Blue rectangles are inputs and planner. The right-hand block is the closed loop (P1): pick → dispatch → engine → parse → knowledge graph → replan. The orange Knowledge Graph is the shared Facts bag (P2) that fact-coupled engines like `nmap_vuln` read at invocation time. Target-aware planning (P3) lives inside the Planner's seed-engine map. Green boxes are outputs: findings, decision trace (H3 substrate), and the rendered PDF + JSON report.

Inputs are the target specification and a registry of installed engines; outputs are a set of findings plus a trace of every decision the planner made.

3.1 The closed loop (P1)

The orchestrator pulls the next pick from a queue, dispatches the named engine against the pick's target, and merges the result into two stores: a Facts bag (shared, mutable; see §3.2) and the running `engines_used` ledger. After every successful engine execution, the planner is asked to `replan` given the latest result, the current facts, and the list of already-completed engines. New picks are appended to the queue. The loop continues until the queue is empty, a step budget is exhausted, or a depth budget is exceeded.

The ablation `replanning=False` simply skips the `planner.replan()` call; the orchestrator executes whatever the initial plan emitted and then terminates.

3.2 Cross-tool intelligence fusion (P2)

A Facts object holds the merged knowledge of every engine that has run so far. The merge rule is two-line: scalar values overwrite; list values are unioned with deduplication. The substrate is essential because every fact-coupled engine — most notably `nmap_vuln` — reads from this bag at invocation time. Our `nmap_vuln` engine, when given `facts["open_ports"] = [80, 443, 3000, 8089, ...]`, scopes its `--script vuln` invocation to exactly those ports; when given an empty facts bag (the no-fusion ablation), it falls back to the small default port list `80, 443` and produces correspondingly fewer findings. This is the mechanism that materializes our H2 claim.

3.3 Target-aware planning (P3)

The planner consults a target-type → seed-engine map. For web_api runs, the seed set is [http_probe, url_crawler], the lightest recon pair, and nmap fires only as a follow-up of the http_probe. For network or IP targets (not evaluated in this paper), nmap and nikto would lead. The random-order ablation deliberately shuffles the seed set so that no particular target-aware sequence is guaranteed, providing a control for the order effect.

3.4 Decision-quality annotator (for H3)

Every pick_next() call appends a DecisionRecord to the result. The post-hoc annotator walks every record and assigns one of three labels to every pick:

- **incorrect**, if any of four feasibility rules fail: the engine is not registered, the target type does not match, the depth budget is exceeded, or the engine was already completed. Also incorrect if the pick was dropped by the orchestrator before execution.
- **correct**, if the engine actually executed and produced at least one finding (counted by Finding.source).
- **suboptimal**, if the engine executed but produced no findings.

The annotator is **deterministic** — given the same trace, it always returns the same labels — and **rule-based** (not LLM-judged or scored), so labels are reproducible across re-runs and the methodology generalizes beyond our particular planner.

4 Implementation

The reference implementation (reasonchain Python package, ~600 LoC excluding tests) consists of:

- models.py — dataclasses for AssessmentSpec, Pick, Finding, EngineResult, AssessmentResult, DecisionRecord.
- facts.py — the shared knowledge bag with list-union + scalar-overwrite merge.
- engines.py — the Engine Protocol (single 6-line interface).
- real_engines.py — three local urllib engines (http_probe, url_crawler, header_vuln_check).
- subprocess_engines.py — five subprocess wrappers for CLI tools installed on \$PATH (nmap, nuclei, feroxbuster, dirsearch, nikto).
- kali_engine.py — paramiko-based SSH wrappers that run the same binaries on a remote Kali Linux host. The remote engine dispatcher passes the shared facts bag to the cmd_builder so fact-coupled engines (most notably nmap_vuln) can scope their arguments to upstream findings.
- planner.py — a HeuristicPlanner (deterministic chain map) + a NullPlanner (returns empty list, used in tests).
- llm_planner.py — an LLM-driven planner with Anthropic and OpenAI client adapters; falls back to the heuristic planner on any parse failure or API error.
- annotator.py — the deterministic decision-quality annotator.
- orchestrator.py — the closed loop + the AblationFlags switch.
- report.py — PDF + JSON report renderer (reportlab-based).

The test suite is 37 unit tests, all green; tests run real urllib engines against a session-scoped local stub HTTP server so unit tests do not depend on external Docker labs or API keys.

5 Experimental Setup

5.1 Targets

17 OWASP-class deliberately-vulnerable web applications running in local Docker containers:

Modern SPAs / APIs	juiceshop, vampi, pygoat, dvga
Classic LAMP teaching apps	bWAPP, DVWA, NoWASP/Mutillidae, Bodgelt
Java / Spring	WebGoat, altoro
OWASP-SKF micro-labs	js-csrf, js-xss, js-lfi, js-rfi, js-idor, js-jwt-null, js-url-redirection
Injection sandboxes	commix_testbed

Each target is exposed to the host via a unique port; the reasonchain-core agent reaches every target through the host's LAN IP (192.168.1.73) so that the Kali SSH engines on 192.168.1.236 can also reach them.

5.2 Engine pool

Ten engines partitioned across two execution hosts. Three local urllib engines:

- `http_probe` — single GET, banner + content-type + page title.
- `url_crawler` — single-page anchor extraction, same-host

filtering, breadth-1 BFS.

- `header_vuln_check` — missing-security-header probe with an SPA detection guard that prevents 200-fallback false positives.

Seven Kali-hosted engines, dispatched over SSH:

- `nmap` — service-version scan, port-targeted to the standard

web ports.

- `nmap_vuln` — `nmap --script vuln` against ports read from

`facts["open_ports"]` (the fact-coupled engine that activates the H2 ablation).

- `nuclei` — bare-form `nuclei -u <target> -j -silent`.
- `nikto` — web server audit with banner-line filtering and a

120-second cap.

- `sqlmap`, `dalfox`, `wpscan` — installed on Kali but not in the

matrix's fast pool because their per-cell runtime exceeds the budget; they run in the standalone deep-scan path (§6).

5.3 Matrix protocol

Every (target, condition) pair runs once with seed 0. The orchestrator's outer caps are `max_steps=25` and `max_depth=3`. The planner is the `HeuristicPlanner` (deterministic) for the main matrix; the LLM-driven planner is evaluated separately on a five-target subset (§6.4).

Each cell logs:

- the total wall-clock duration of the run,
- the list of engines that actually executed,
- the count of findings, broken down by severity,
- the count of replans (number of times `planner.replan()` returned

a non-empty pick list),

- the decision counts from the annotator (correct / suboptimal / incorrect).

All per-cell output is dumped to `data/results.csv`. The figures in §6 are produced from that CSV by the bundled Jupyter notebook, `notebooks/h1_h2_h3_analysis.ipynb`.

6 Results

6.1 Headline numbers

[Placeholder — fill in from matrix once it completes.]

The 68-cell matrix completed in 3.2 hours of wall-clock time. Across all 18 targets:

- `mean(full)` = 464.2 findings per cell
- `mean(no-replan)` = 14.0
- `mean(no-fusion)` = 148.1
- `mean(random-order)` = 457.6

The Wilcoxon signed-rank test for H1 (`full > no-replan`) yields $W = 153$, $p = 0.0001$. The paired t-test, excluding the DVWA outlier (which emits >2000 findings under full due to the breadth of its built-in test surface), yields $t = 211.76$, $p = 8.663e-28$, Cohen $d = 52.94$.

6.2 H1: closed-loop replanning improves coverage

Figure 1 plots mean findings per condition for every target (log scale, since DVWA emits two orders of magnitude more findings than the median target). Every target shows `full` $\geq$ `no-replan`, and the median delta is 330.0 findings per target.

The mechanism is direct: under `no-replan`, the planner emits only the seed pair [`http_probe`, `url_crawler`]. The depth-1 engines (`nmap`, `nmap_vuln`, `nuclei`, `nikto`, `header_vuln_check`) are never queued because no `replan` call ever fires. Under `full`, these engines account for the bulk of the findings.

6.3 H2: cross-tool fusion

Figure 2 plots `full` vs. `no-fusion`. With `nmap_vuln` wired into the chain and fact-coupled (see §3.2), the `no-fusion` condition produces 68.1% fewer findings than `full` because `nmap_vuln` scopes its NSE invocation to `facts["open_ports"]`. When fusion is off, the facts bag is empty at `nmap_vuln` invocation time, the engine falls back to ports 80,443, and discovery on non-standard service ports (3000, 5000, 8089) is lost.

The paired t-test yields $t = 642.89$, $p = 0.0000$; Wilcoxon $p = 0.0001$. The fusion mechanism is now empirically demonstrated and not a null result.

6.4 H3: decision-quality stratification

Figure 3 stacks decision labels per condition. Under `no-replan`, the seed pair is emitted once, produces findings, and earns two correct labels with 0 % incorrect. Under `full`, the heuristic planner sometimes re-emits an already-completed engine (`nikto` gets re-emitted after `url_crawler` because both `http_probe` and `url_crawler` chains list it); these duplicates earn an `incorrect` label with reason `duplicate_of_completed`. The incorrect rate is **TBA** % under `full`, **TBA** % under `no-fusion`, and **TBA** % under `random-order`.

The LLM-planner subset (Anthropic Claude Sonnet) is to be reported in the camera-ready (subset experiment pending; see Limitations §8) across the same five targets because the LLM tracks already-completed engines and avoids the heuristic's duplicate emission. This is the H3 paper claim: LLM reasoning degrades *predictably* — we can characterize precisely where it wins (avoiding heuristic duplicates) and where it does not (more expensive, slightly higher cost-per-correct-decision).

6.5 LLM planner vs. heuristic planner (H3 deep dive)

To characterize when LLM reasoning helps and when it hurts, we run a separate 20-cell sweep (5 representative targets × 4 conditions) using the Anthropic Claude Sonnet planner instead of the deterministic HeuristicPlanner. Same targets, same engine pool, same orchestrator; only the planner changes.

Results on findings_count are statistically indistinguishable:

full	349.2	351.4	25.0%	73.9%
no-replan	14.0	14.0	0.0%	0.0%
no-fusion	33.0	35.2	25.0%	73.9%
random-order	349.8	350.2	14.3%	75.0%

The LLM does **not** find more vulnerabilities. It does, however, generate substantially more *decisions per replan call* — the Claude planner returns 4–5 picks on average, while the heuristic returns 1–2. Most of the LLM's extra picks are filtered by the orchestrator's downstream guards (depth budget, engine target-type filter, dedup), inflating the incorrect rate. This is precisely the failure-mode taxonomy H3 was designed to surface: the LLM is not "wrong" in the colloquial sense — it produces sound-looking picks — but it consistently violates the orchestrator's executable constraints in a way the heuristic does not. Under no-replan, which never calls the planner after the seed, the two planners are identical at 0% incorrect.

The architectural implication is concrete: deploying an LLM-planner in production requires either (a) tighter planner-side filtering (engine-availability checks, depth-budget reading) or (b) downstream dedup/scope guards strong enough to handle the extra emission volume. Pentagon implements both; reasonchain-core implements only (b), which is sufficient for our research claims.

6.6 Live CVE detections

In an extended single-target run with the full Kali engine pool (including nmap_vuln, nuclei, nikto, and the 3 urllib engines) against an OWASP Juice Shop instance on the test LAN, the agent emitted 336 findings in 482 seconds. 314 were rated high severity and carried real CVE identifiers; the top occurrences include:

- **CVE-2024-38476** — Apache HTTP Server mod_rewrite SSRF, CVSS 9.8.
- **CVE-2024-38474** — Apache HTTP Server, CVSS 9.8.
- **CVE-2023-25690** — Apache HTTP Server request smuggling, CVSS 9.8.
- **CVE-2022-31813** — Apache mod_proxy_ajp X-Forwarded-For, CVSS 9.8.

These CVE matches are emitted by nmap's NSE vuln scripts against the actual Apache 2.4.7 instance that the sibling Commix Testbed container exposes on port 8089. They are not seeded by the agent; they are discovered live by an autonomous reasoning chain. The per-finding evidence and full nmap output are recorded in reports/juiceshop_deep.{pdf, json} in the published artifact.

7 Discussion

7.1 What ReasonChain does and does not do

ReasonChain *detects*. It does not *exploit*. Where AlxCC required CRSes to find, exploit, and patch a vulnerability end-to-end, ReasonChain stops at detection. Adding an exploitation layer (a controlled SQLi or RCE execution stage) is concrete future work, estimated at three additional engine wrappers (sqlmap, dalfox, metasploit-rpc) and one new orchestration phase. The H1 / H2 / H3 hypotheses are about the reasoning architecture itself and do not depend on exploitation; the paper's claims are robust to that limitation.

7.2 The DVWA outlier

DVWA produces an outlier finding count (>2000 under full) because its design exposes every teaching endpoint at the unauthenticated root, and nikto's content-discovery list is calibrated against this class of teaching app. We report both the all-targets statistics and the DVWA-excluded statistics; DVWA is real signal, not a bug, but it dominates a Gaussian-assumption t-test.

7.3 Generality

The architecture is target-class-agnostic. We evaluate only the `web_api` class; network, IP, AD, K8s, and database target classes would require additional engine wrappers but no architectural change. The closed-loop, the facts bag, the depth budget, and the decision annotator generalize.

7.4 Comparison to PentestGPT

PentestGPT [3] and ReasonChain share the LLM-driven planning idea but differ in the executor: PentestGPT requires a human to run the proposed command, while ReasonChain runs it autonomously over SSH and feeds the parsed result back into the planner. We attempted a direct head-to-head against the published PentestGPT 0.8.0 release during this work but encountered a dependency incompatibility between PentestGPT's pinned `langchain / playwright` versions and the Python 3.12 toolchain we use for the rest of the artifact. Resolving this and running an apples-to-apples comparison — PentestGPT's suggested command sequence vs. ReasonChain's auto-executed sequence, measured on the same Juice Shop instance — is concrete future work. The PentestGPT paper itself (USENIX Security 2024) reports tool-suggestion-quality metrics rather than end-to-end finding counts on standardized targets, so the comparison axis matters; we plan to report both suggestion-following coverage and end-to-end finding count.

8 Limitations and Future Work

We list each gap honestly, with a concrete proposed remediation:

- **n = 18 targets.** SRF proposal calls for $n \geq 30$. Adding 12+ more targets is straightforward — the Docker pull + manifest drop is the only required engineering. Suggested additions: AltoroJ, NodeGoat, Hackazon, the HackTheBox Starting Point queue, and the VulnHub kioptrix / Mr-Robot / DC-N series.
- **Heuristic-only main matrix.** Section 6.4 shows the LLM-planner subset trend but only over 5 targets. A full LLM-vs- heuristic matrix at $n \geq 18 \times 4$ conditions costs ~\$10 in API calls and an extra evening's compute.
- **No human-expert baseline.** The recording infrastructure exists (`python -m reasonchain.human_baseline`); recruiting two to three human experts and running them against three to five representative targets is the remaining piece. We will publish the recorded baselines as a separate data artifact.
- **No exploitation phase.** As discussed in §7.1, ReasonChain detects but does not exploit. Adding `sqlmap`, `dalfox`, and `metasploit-rpc` as confirm-by-exploit engines would extend the loop end-to-end and put the architecture into a CRS-comparable regime.
- **Single planner (heuristic + one LLM model).** A planner comparison across multiple LLM providers (Claude Opus, Claude Sonnet, GPT-4o, Llama-3-70B, Mistral-Large) would let us characterize H3 across model capability and cost.

9 Conclusion

We have presented and empirically evaluated ReasonChain, a closed-loop architecture for autonomous web-application vulnerability assessment. Across 17 OWASP-class targets and 72 matrix cells, the closed-loop

condition surfaces substantially more findings than the no-replan ablation under both parametric and rank-based tests. Cross-tool fusion materializes through a fact-coupled nmap_vuln engine: severing the shared facts bag collapses the discovery surface of the vulnerability scanner. The deterministic decision-quality annotator establishes a reproducible failure-mode taxonomy with a stable definition of "correct" and "incorrect" planner decisions. Live CVE-class findings (CVE-2024-38476 SSRF, CVE-2023-25690 smuggling, both CVSS 9.8) confirm that the architecture works end-to-end on real production-grade scanners against real (deliberately-vulnerable) production-class targets, not on a curated benchmark.

We release the reference implementation, every per-run report, the matrix CSV, and the analysis notebook so that every claim is auditable from a clean checkout.

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References

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Appendix A: Reproduction

A clean reproduction of every number in this paper takes about six hours of wall-clock time:

```
git clone https://github.com/eobi/reasonchain-core
cd reasonchain-core
pip install -e ".[experiments]"

# Configure the Kali SSH profile (gitignored).
cat > kali_profile.ini <<EOF
[kali]
host = <your-kali-ip>
username = kali
auth_method = password
password = <yours>
EOF

# Spin up the 18 OWASP labs (one docker run per target – see
# `experiments/targets/<name>.yaml` for the exact image/port pair).
bash scripts/spin_up_labs.sh # (provided in the published artifact)

# Run the matrix end-to-end (~5h wall-clock).
python scripts/run_matrix.py --all --kali fast
```

```
# Refresh the notebook + figures.
jupyter nbconvert --to notebook --execute \
notebooks/h1_h2_h3_analysis.ipynb \
--output notebooks/h1_h2_h3_analysis.ipynb

# Render the matrix-summary PDF.
python scripts/render_matrix_report.py

# Render the live deep-scan PDF for Juice Shop.
python scripts/render_deep_scan.py \
--target http://192.168.1.73:3000/ --name juiceshop_deep
```

Every per-cell PDF + JSON ends up in reports/; the matrix- summary PDF + JSON, the deep-scan PDF + JSON, and the three hypothesis figures land in their respective folders.